REPORT — Bridge C order4a (explicit shadow candidate)

PREREG sha256 cb914b77…ed34b (unchanged). Script hash in scripts/SHA256SUMS. Category B: numerical ranks. Mod-p certification is not feasible because the entries are transcendental.

Sign control (gating)

- A0 at w=1: rank 36, closed.
- Wrong sign: joint rank 72 > 36.
- **Passed.**

Tests: A1^c = A1 − Σ ε_Q W[Q]

ε (W[1],W[7/2],W[595/2])	rank	joint w=½	joint w=1	closed
000 (A1)	105	210	210	no
001	91	182	182	no
010	104	208	208	no
011	90	180	180	no
100	105	210	210	no
101	87	170	156	no
110	103	206	206	no
111 (primary: D+R)	58	116	98	**no**

Verdicts

- **Primary:** candidate falsified (Category B). D+R does not close at w=½ or at w=1.
- **Secondary:** no ε closes.

order4a notes (descriptive)

- **Rank 58.** D+R has rank 58, the same as D alone (order3). In this column space, removing all three sawtooth groups leaves a rank-58 object whose S-image still adds 40–58 new directions. The fixed-coefficient sawtooth candidate does not absorb the defect.
- **Untested variant.** The preregistered family used coefficients ε_Q ∈ {0,1}. A completion with non-trivial, analytically derived coefficients was not tested: for example, erf-completion constants that depend on the angle between the wall and the null directions of the indefinite form. That is the natural refinement. It needs the constants derived from the actual completion of H_7 first, then preregistered.
- **Category.** Category B only: numerical ranks at threshold 1e-9 over 216 points.